Mattias Eek

M.Sc. in Engineering, Robotics

mattiaseek12@outlook.com | +46 70 735 92 32 | linkedin.com/in/mattias-eek-16893a26a

Professional Profile

M.Sc. in Engineering, Robotics, from Mälardalen University (2026). Robotics engineer with hands-on experience in control systems, embedded software, robotics integration, electronics, sensors and actuators, and applied machine learning. Delivered multidisciplinary engineering projects spanning LiDAR stabilisation, ROS2 robotics, Bluetooth localisation, and simulation-driven control design.

Technical Skills

Programming: Python, C, C++, MATLAB, Zephyr

Robotics: ROS 2, Simulink, Simscape Multibody

CAD: SolidWorks, Onshape

Tools and platforms: Linux, Visual Studio Code, FoxGlove, LabVIEW, Multisim, Ultiboard, Wegstr CNC

Education

M.Sc. in Engineering, Robotics: Mälardalen University, 2026

Engineering Projects

Active Gimbal Stabilisation: Mälardalen University, SLaRC

Improving Long-Range LiDAR Measurement Accuracy Through Active Gimbal Stabilisation in Dynamic Environments

- Designed a two-degree-of-freedom active gimbal control system to stabilise a Livox MID-360 LiDAR sensor on an unmanned ground vehicle.
- Built and validated MATLAB/Simulink + Simscape Multibody models with PID control.
- Evaluated controller performance using ROS2 bag recordings and real sensor data.
- Validated system performance using Python-based point-cloud analysis (RANSAC, ICP, C2C, and M3C2).

Unmanned Ground Vehicle: Mälardalen University, SLaRC

- Developed motor-control software, including CAN communication, for a four-wheel-drive ROS 2-based UGV in C++.
- Integrated motor-control hardware and validated real-time operation and teleoperation functionality using ROS 2.
- Contributed to mechanical design in SolidWorks, hardware integration, and system-level testing.

Bluetooth Indoor Localisation with Machine Learning: Mälardalen University

- Developed a BLE RSSI fingerprinting localisation system in Python using KNN and SVM.
- Improved localisation accuracy through data processing, interpolation and feature selection.

Selected Additional Engineering Projects:

- Motor-Controlled LiDAR System – Designed a motor-controlled LiDAR system including PCB design, H-bridge motor driver, LabVIEW control software, and sensor calibration
- ECG Signal Processing – Developed MATLAB algorithms using wavelet transforms and heart-rate variability analysis to process ECG data from elite athletes.
- RL Navigation Stack – Developed a hierarchical reinforcement-learning navigation stack using ROS 2 and Gazebo, with PPO for subgoal selection, obstacle avoidance and differential-drive control.
- ChessAI – Built a chess engine in C++ with alpha-beta search, iterative deepening and a PyTorch-trained neural evaluator, optimized for native inference and benchmarked against Stockfish 19.

Languages

- **Swedish:** Native
- **English:** Fluent